

 CSMSS Chhatrapati Shahu Maharaj Shikshan Sanstha's CHH. SHAHU COLLEGE OF ENGINEERING Kanchanwadi, Paithan Road, Chhatrapati Sambhajinagar 431 011 	
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE	ASSIGNMENT

ACADEMIC YEAR	2025-26	COURSE TEACHER	
COURSE NAME	P&S	CLASS	S.Y.
SEM.	IV	DATE	21/04/2026

ASSIGNMENT NO. 02

Que. No.	Question Description	Marks	CO No.	BT Level																		
1)	A random variable X has the following distribution: Determine (i) k (ii) $P(X < 4)$ (iii) $P(2 \leq X < 5)$ (iv) $E(x)$ (v) $Var(X)$ <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$p(x)$</td><td>2k</td><td>4k</td><td>6k</td><td>8k</td><td>10k</td><td>12k</td></tr></table>	x	1	2	3	4	5	6	$p(x)$	2k	4k	6k	8k	10k	12k	5	CO2	L3				
x	1	2	3	4	5	6																
$p(x)$	2k	4k	6k	8k	10k	12k																
2)	Define random variable. Suppose a random variable X has the following probability distribution function then find the probability that: (a) X is at least 10 (b) X is more than 10 (c) X is less than 4 (d) X is at least 4 and at most 9 <table><tr><td>x</td><td>1</td><td>3</td><td>4</td><td>7</td><td>9</td><td>10</td><td>14</td><td>18</td></tr><tr><td>$p(x)$</td><td>0.11</td><td>0.07</td><td>0.13</td><td>0.28</td><td>0.18</td><td>0.05</td><td>0.12</td><td>0.06</td></tr></table>	x	1	3	4	7	9	10	14	18	$p(x)$	0.11	0.07	0.13	0.28	0.18	0.05	0.12	0.06	5	CO2	L3
x	1	3	4	7	9	10	14	18														
$p(x)$	0.11	0.07	0.13	0.28	0.18	0.05	0.12	0.06														
3)	In a recent little league softball game, each player went to bat 4 times. The number of 6 hits made by each player is described by the following probability distribution. What is mean and standard deviation of the probability distribution. Find $P(X \leq 2)$ and $P(X \geq 2)$ <table><tr><td>No. of hits x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Probability $p(x)$</td><td>0.10</td><td>0.20</td><td>0.30</td><td>0.25</td><td>0.15</td></tr></table>	No. of hits x	0	1	2	3	4	Probability $p(x)$	0.10	0.20	0.30	0.25	0.15	5	CO2	L3						
No. of hits x	0	1	2	3	4																	
Probability $p(x)$	0.10	0.20	0.30	0.25	0.15																	
4)	Obtain the probability distribution of X, the number of heads in three tosses of coin. Also find expected number of heads appearing when a fair coin is tossed three times.	5	CO2	L3																		
5)	Four bad apples are mixed accidentally with 20 goods apples. Obtain the probability distribution of the number of bad apples in a draw of 2 apples at random.	5	CO2	L3																		
6)	Find the Mean, Variance and Standard Deviation of the discrete random variable x whose probability distribution is given as <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$P(X)$</td><td>0.1</td><td>0.1</td><td>0.3</td><td>0.3</td><td>0.2</td></tr></table>	X	1	2	3	4	5	$P(X)$	0.1	0.1	0.3	0.3	0.2	5	CO2	L3						
X	1	2	3	4	5																	
$P(X)$	0.1	0.1	0.3	0.3	0.2																	
8)	Define a random variable. Explain the mean and variance along with the properties.	5	CO2	L2																		
9)	Suppose X and Y represent the operating lives of bulbs (X) and Tubes (Y) in a months. Their joint probability density function is given by $f_{X,Y}(x,y) = \begin{cases} e^{-x-y}; & x \geq 0, y \geq 0 \\ 0; & \text{otherwise} \end{cases}$ Find i) The probability of bulbs has life less than or equal to 1 month. ii) The probability of Tubes has life more than 2 months.	5	CO2	L3																		
10)	If (X,Y) be Pair of discrete random variables each taking three values 1, 2 and 3 with the joint probability distribution as given in tables. <table><tr><td>$X \rightarrow$ $Y \downarrow$</td><td>1</td><td>2</td><td>3</td></tr><tr><td>1</td><td>5/27</td><td>4/27</td><td>2/27</td></tr><tr><td>2</td><td>1/27</td><td>3/27</td><td>3/27</td></tr><tr><td>3</td><td>3/27</td><td>4/27</td><td>2/27</td></tr></table> Obtain the marginal probability distribution of X & Y and hence find $E(X)$, $E(Y)$ & $E(X + Y)$. Also find $Var(X)$ and $Var(Y)$.	$X \rightarrow$ $Y \downarrow$	1	2	3	1	5/27	4/27	2/27	2	1/27	3/27	3/27	3	3/27	4/27	2/27	5	CO2	L3		
$X \rightarrow$ $Y \downarrow$	1	2	3																			
1	5/27	4/27	2/27																			
2	1/27	3/27	3/27																			
3	3/27	4/27	2/27																			

11)	The probability distribution of random variable X is given in the table				5	CO2	L3
	x	-2	1	3			
	$p(x)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$			
	Find $E(X)$, $E(2X + 5)$ Also find $E(X^2)$						
12)	Find the Mean, Variance and Standard Deviation of the discrete random variable x whose probability distribution is given as				5	CO2	L3
	X	1	2	3			
	$P(X)$	0.1	0.1	0.3			
13)	Find the variance of the number obtained on a throw of an unbiased die.				5	CO2	L3
14)	A die is thrown twice. Getting 'a number greater than 4' is considered a success. Find the mean and variance of the probability distribution of number of successes.				5	CO2	L3